

OVERVIEW

Having conversations about the real world in your math classroom, means opening up to all of the related and connecting thoughts your students may wish to share on the topic. Have your students ever pulled your lesson into a conversational tangent that ate up all of your class time and best intentions? In this conversation, you'll have an opportunity to explore how to navigate conversational rabbit holes while still honoring student questions and contributions about a real world topic.

SPARK MEDIA & FOCUS LESSON

Classroom Dialogue

A class has a lively launch discussion, in the lesson [Big Foot Conspiracy](#).

PEDAGOGICAL THEME

Focusing on the Real World

USING THE CONVO GUIDE FOR A GREAT ROUNDTABLE DISCUSSION

- This conversation is designed for a group of 2-5 educators over 30-60 minutes.
- The Convo Guide serves as an agenda for a roundtable conversation where everyone is there both to learn and to share their expertise.
- Each participant should have their own copy of the Convo Guide and access to the Spark Media.
- Pace yourselves to get through the entire agenda in the time you have.
- Questions aren't intended to have a single right answer; instead, they're designed to spark rich discussions.
- At the end, everyone will commit to trying "one small thing" the very next day. Consider how you might follow up with one another!

OBSERVE

5–15 MIN

- To start, take a look at the Spark Media.
- Which student comments show that they're activating prior experiences or knowledge relating to this lesson? What are some comments that derail or detract from the conversation?
- How do some of the teacher's responses support or hinder the following goals they have for this lesson launch?

DISCUSS

10–15 MIN

- Conversations about the real-world can elicit unexpected comments that have the potential to veer the discussion in unexpected directions. What are some different types of rabbit holes that you may encounter while talking about the world? How do you decide whether to reign in a conversation or to go with the flow?
- When a teacher does want to redirect the conversation, what are some ways they can move on while still honoring the student who shared the comment or question?

RELATE

10–15 MIN

- Consider a recent time the conversation in your own classroom slipped down a rabbit hole (for better or worse). How did this impact the lesson?
- What new moves would you like to explore for navigating around or through rabbit holes in your next real world conversation? What will you need to do while preparing a lesson to prepare to try this?

...AND COMMIT!

5–10 MIN

- What is one small thing you can do tomorrow in support of those shifts?

A PEEK INTO MR. ALICE'S CLASSROOM

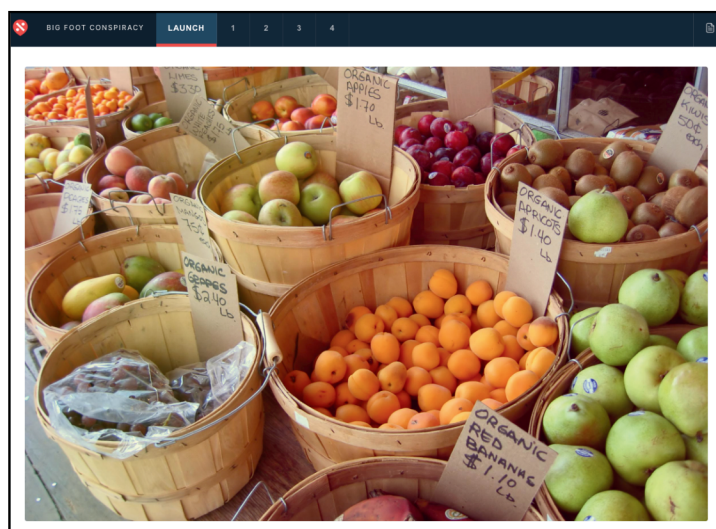
The following transcript offers a glimpse into the first few minutes of Mr. Alice's classroom as they are beginning the Citizen Math lesson Big Foot Conspiracy which explores the question: Should people with small feet pay less for shoes?

Each Citizen Math lesson begins with 5 minutes for a brief "Launch" to get the conversation started. The goals of this time are (1) to break the ice and get students talking, (2) to allow students to begin to personally connect this topic to their world, and (3) to establish some key point that will motivate the lesson.

What's projected in the Launch

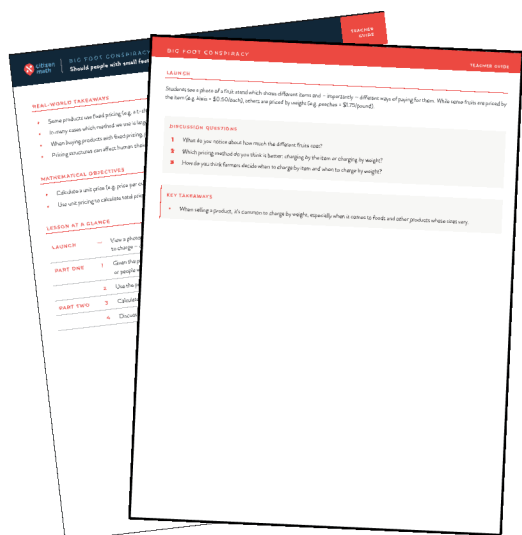
In Teach Mode for Big Foot Conspiracy, the Launch offers a photo of a fruit stand which shows different items and – importantly – different ways of pricing them.

While some fruits are priced by the item (e.g. kiwis = \$0.50/each), others are priced by weight (e.g. peaches = \$1.75/pound).



What's in the Teacher Guide

The Teacher Guide offers up these discussion questions to help get the conversation going and establish the Key Point of the launch.



Discussion prompts:

- What do you notice about how much the different fruits cost?
- Which pricing method do you think is better: charging by the item or charging by weight?
- How do you think farmers decide when to charge by item and when to charge by weight?

Key Point to establish in the Launch:

"When selling a product, it's common to charge by weight, especially when it comes to foods and other products with sizes that vary."

CLASSROOM DISCUSSION TRANSCRIPT



MR. ALICE

STUDENTS

Alright class, today we are going to talk about pricing. What do you notice about the fruit in this photo?

Oh, cool, the farmer's market?

By a show of hands, Who else has been to a farmers' market...even if it's a different one than Ina goes to?

Okay, and by a show of hands, now who has shopped or helped someone else shop at a grocery store?

Ina:

Ooh! The market my family goes to uses baskets like those!

Ina:

Yes. On Saturdays.

[about half of the students raise hands]

[almost all of the students raise hands]

MR. ALICE

STUDENTS

Okay. Let's look specifically at the pricing in this photo to see if it matches what you saw at the grocery store or farmer's market. Take a look at the sign for mangos. What does that sign say?

And, Jose, the price for Kiwis?

We're only talking about kiwi the fruit right now, friends.

What about the grapes, Emeril?

Marcus, Tell me what that means: "per pound".

Marcus:

Seventy five cents.

Ina:

Did you know kiwis are also birds? They can't fly and they have these really long beaks.

Jose:

AND people used to think they used their beaks for fighting but in actuality....

Jose:

Sorry. Yeah, those are fifty cents each.

Emeril:

Two dollars and forty cents. "Ell bee?"

Marcus:

LB is pound. It's per pound.

Marcus:

You have to weigh them. So when you check out, they'll put them on a scale to figure out how much they weigh and that tells you how much they cost.

MR. ALICE

STUDENTS

Yeah, my grocery store does the same thing. I think most stores do that where at the checkout there's a scale in front of the register for produce.

That's ok. But do you ever buy meat or cheese from the deli...it's usually in the back of the grocery store, where there's the big glass case with a whole bunch of meats and cheeses, and there's someone's behind the counter who will slice however much you want?

That's priced by the pound, too..

Would anyone like to mention anything else in the grocery store they've seen priced "by the pound?"

Yeah, and does it make sense to you that a big heavy turkey costs more than a little bitty turkey?

I did! Alright, so we know some things are priced by the pound and some things are priced by the item. Do you think one way is better than another?

.....

How do you think the farmers at the farmers market decide which way to price their fruit?

Bobby:

It all looks gross to me. I don't eat fruit and don't get me started on veggies.

Bobby:

Oh right!

Clara:

Oh, a turkey! We got a HUGE turkey for thanksgiving, and they're so expensive.

Clara:

Sure! I mean, you're getting more turkey.

Xavier:

And also tomatoes are sold by weight. Did you know that tomatoes are actually a fruit?

MR. ALICE

STUDENTS

Xavier:

Well, if they're hard to count like grapes, maybe it's easier to just weigh them. But with something big, I think it's easier to just count them. If it's off by a little bit, whatever....

[Pause]

...It's not like it took any more work to grow a little mango than a big mango.

Ina:

But if something's a lot bigger...like if they're selling watermelons and some of them are huge and some of them are small, then they'd probably want to charge more for the bigger watermelon. Because they can. And because people aren't going to pay all that much for a little watermelon that's only good for like two people.

Ah! Okay, so I hear you saying that you think the farmer should charge more for a big watermelon than for a small one, because the person getting the small one just isn't getting as much watermelon. That makes sense to me.

It sounds like it would make sense for a lot of things. What if other things that come in big and small sizes were priced by the pound. What if the price changed depending on the weight for something like, say, SHOES?!

Well let's dig into it...

Bobby:

That'd be crazy is what if!